

Research Checkpoint: Toward a Universal Semantic Compiler for CNF Satisfiability

Exact Boundary Relations,
Emergent Normal Forms, Saturation, and Global Concurrency

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August 2026

Abstract

This note records an experimental and mathematical checkpoint in the development of a universal semantic approach to Boolean satisfiability.

The starting point is the exact compatibility formulation in which every CNF formula is represented by locally admissible relations connected through shared Boolean coordinates. The present investigation asks a stronger question: whether the explicit recognition grammar of specialized semantic charts can be replaced by a single family-independent compiler acting directly on arbitrary finite Boolean relations.

The current experimental evidence supports a four-stage architecture:

exact relation discovery $\longrightarrow$ canonical compression $\longrightarrow$ global semantic saturation $\longrightarrow$ global concurrency.

No benchmark-family names are required in the experimental pathways described here. Structures previously handled by distinct recognizers—categorical EXACT1 covers, trimmed parity systems, parity-noise systems, and alternating-cycle macrostates—re-emerge from general operations on exact boundary relations.

Five structurally different benchmark instances have currently been recovered to full SAT by experimental family-independent pathways: UF250, ac-sgen, two-trees, MDP, and X9. Battleship is presently reduced to one remaining unsatisfied clause by the corresponding generic pathway and is therefore retained as an active counterexample rather than treated by residual repair.

The official DREAM6 v186 implementation remains unchanged. All results reported here belong to a separate experimental branch. No universal completeness claim is made.

1 Objective

The previous semantic framework uses several mathematically exact or certified representations:

OR, EXACT1, GF(2), BDD, APN, EVEN_CYCLE,

together with deterministic global section recovery.

These representations are useful, but a fundamental question remains:

Can the recognition grammar itself be replaced by one universal rule acting on arbitrary CNF formulas?

The target is deliberately not phrased as a complexity-class statement. The immediate problem is constructive:

derive the relevant semantic structure directly from the formula, without naming its encoding family.

The weaker structural theorems can be extracted later if the universal construction eventually fails. The present checkpoint records the direct attack on the general rule.

2 Exact boundary semantics

Let

$$F = H \wedge G$$

where H is a subformula. Define its internal and boundary variables by

$$I(H) = \text{Var}(H) \setminus \text{Var}(G),$$

$$\partial H = \text{Var}(H) \cap \text{Var}(G).$$

The exact semantic meaning of H as seen from the rest of the formula is

$$R_H = \left\{ y \in \{0, 1\}^{\partial H} : \exists z \in \{0, 1\}^{I(H)} \quad H(y, z) = 1 \right\}.$$

Theorem 2.1 (Exact boundary replacement). *Let $F = H \wedge G$, where H and G interact only through ∂H . Then*

$$F \text{ is satisfiable} \iff R_H \wedge G \text{ is satisfiable.}$$

Proof. If (y, z, w) satisfies $H \wedge G$, then $y \in R_H$, hence (y, w) satisfies $R_H \wedge G$.

Conversely, if (y, w) satisfies $R_H \wedge G$, the definition of R_H guarantees some z such that $H(y, z) = 1$. Thus (y, z, w) satisfies $H \wedge G$. $\square$

This immediately gives a complete representational grammar:

$$\boxed{R_{H_1 \cup H_2} = \exists u, (R_{H_1} \bowtie R_{H_2}).}$$

The problem is therefore not the existence of exact semantic relations. The problem is whether they admit compact and operationally useful representations throughout the compilation.

3 Arbitrary finite relations as the primitive language

The experimental compiler abandons semantic family labels at its lowest level.

A semantic atom is simply a finite Boolean relation

$$R \subseteq \{0, 1\}^S$$

on support S .

The current prototype represents such relations primarily by reduced ordered binary decision diagrams (ROBDDs), although the theory does not require BDDs to be the final universal representation.

The important operations are:

$$R_1, R_2 \longmapsto R_1 \wedge R_2,$$

$$R \longmapsto \exists x, R,$$

$$R \longmapsto R|_{\phi},$$

where ϕ is a globally certified invariant or boundary restriction.

Every accepted transformation is exact unless explicitly classified as a certified selector.

4 Why maximal exact elimination is insufficient

A direct exact attack uses Davis–Putnam elimination:

$$R_{\text{new}} = \exists x \bigwedge_{R \ni x} R.$$

This is family-independent, exact, and complete.

It nevertheless exposes the principal compression barrier.

On UF250, explicit resolution remains manageable for an initial segment of the elimination order but then grows rapidly. In the experimental min-boundary/min-fill probes, the number of intermediate resolvents eventually increases by orders of magnitude.

Replacing explicit clauses by one global ROBDD significantly delays this explosion but does not remove it. The induced boundary width remains too large.

This yields the first negative result of the present checkpoint:

universal exact elimination is complete but not automatically compact.

The universal compiler must therefore preserve factorization whenever global elimination is semantically exact but geometrically destructive.

5 Factorized exact compression

Instead of forcing

$$F \longrightarrow R_F$$

as one monolithic object, maintain

$$F \equiv \bigwedge_{j=1}^k R_j,$$

where each R_j is an exact finite relation.

Candidate neighboring factors are joined only when canonicalization produces a sufficiently compact relation.

This gives the first universal architectural principle:

compress where cheap, factorize where not.

There are then two conceptually different exact representations of the same formula:

1. an *orientation representation*, optimized for smooth or partition-based information transport;
2. a *section representation*, optimized for exact global concurrence and proof-carrying reconstruction.

They need not use identical factorization.

This distinction turned out to be essential for UF250.

6 Universal arbitrary-relation partition operator

For a factor

$$R_f \subseteq \{0, 1\}^{S_f}$$

with incoming log-fields L_i , define

$$Z_f = \sum_{x \in R_f} \exp \left(\sum_{i \in S_f} L_i x_i \right).$$

For each variable $i \in S_f$, define conditioned partition functions

$$Z_f^{(i=0)} = \sum_{\substack{x \in R_f \\ x_i=0}} e^{\sum_j L_j x_j},$$

$$Z_f^{(i=1)} = \sum_{\substack{x \in R_f \\ x_i=1}} e^{\sum_j L_j x_j}.$$

The exact local response is

$$U_{f \rightarrow i} = \log Z_f^{(i=1)} - \log Z_f^{(i=0)}.$$

For an arbitrary BDD relation, all such values can be evaluated by forward/backward weighted model counting over the decision diagram.

Thus OR, EXACT1, parity-derived relations, capacity relations, and arbitrary small Boolean relations can all share one local operator.

The current experimental orientation schedule uses

$$N = 200, \quad \alpha = 0.2, \quad \rho(t) = 0.4 \left(\frac{t}{N} \right)^2$$

in the successful UF250 and X9 experiments.

These numerical values are experimental parameters, not theorem-level constants.

7 Universal hard concurrence

For exact factors

$$R_1, \dots, R_k,$$

introduce one local copy of a Boolean variable for every factor incidence.

Define

$$A = \prod_f R_f,$$

the direct product of exact local relation sets.

Define

$$B = \{\text{all local copies of each Boolean variable agree}\}.$$

Then

$$A \cap B \cong \text{Sat}(F)$$

provided the exact factor cover represents the complete original formula.

Projection onto A is performed independently factor by factor. For BDD relations it is computed by dynamic programming for the nearest accepted Boolean state.

Projection onto B is simply copywise averaging.

A difference map with fixed experimental parameter

$$\beta = 0.55$$

is then used as a generic global concurrence mechanism.

Importantly, hard concurrence is not always a good orientation mechanism. Several successful experiments required exact partition transport first and hard concurrence only as the final section-recovery stage.

This motivates the general decomposition

$$\boxed{\text{soft exact transport} \longrightarrow \text{hard exact concurrence.}}$$

8 Generic semantic atom discovery

Several family-independent discovery rules have emerged.

8.1 Same-support atomicity

If several clauses have exactly the same variable support S , compile their conjunction into the exact relation

$$R_S = \bigcap_{\text{supp}(C)=S} C.$$

Accept it as an atom when canonical representation is significantly more compact than the collection of raw clauses.

This simple rule automatically produced:

510

minimal atoms for two-trees, and

380

minimal atoms for MDP.

No parity syntax was queried.

8.2 Support closure

For a candidate support S , define

$$H(S) = \{C : \text{Var}(C) \subseteq S\}.$$

Compile

$$R_S = \text{Sat}(H(S))$$

exactly.

A region is accepted if the relation is compact relative to the constraints absorbed into it.

Applied to ac-sgen, this rule recovered exactly

36

five-variable, ten-clause compressed regions without testing for pairwise AMO or EXACT1 syntax.

8.3 Low-arity interaction components

Construct the interaction graph induced by homogeneous low-arity constraints.

Connected components are candidate semantic regions. They are accepted only if their exact relation strongly compresses.

On Battleship, this yielded

26

disjoint components, each containing

14 variables, 91 binary clauses,

whose conjunction has only

15

allowed states.

The compiler does not need to call this an AMO relation.

8.4 Strong-overlap contraction

Once exact atoms have been discovered, build their overlap graph.

A general emergent principle observed on X9 is:

internalize multi-variable overlaps; retain singleton overlaps for global concurrence.

The resulting connected components may then be joined and canonicalized into higher-level semantic macrorelations.

On X9 this rule produced exactly

120

macrocomponents of four local regions each, without detecting cycles syntactically.

9 Global semantic saturation

Compression alone is insufficient. Exact local relations may contain semantic information that becomes stronger only after coupling to the rest of the formula.

The compiler therefore needs a saturation layer.

The experimental evidence currently reveals at least three general types of saturation.

9.1 Hamming-weight saturation

For relation

$$R_f \subseteq \{0, 1\}^{S_f},$$

define its exact local Hamming interval

$$\ell_f = \min_{x \in R_f} \sum_{i \in S_f} x_i,$$

$$u_f = \max_{x \in R_f} \sum_{i \in S_f} x_i.$$

These give valid linear inequalities

$$\ell_f \leq \sum_{i \in S_f} x_i \leq u_f.$$

Combine these inequalities globally over the shared Boolean variables.
If global optimization proves

$$\min \sum_{i \in S_f} x_i = \max \sum_{i \in S_f} x_i = k,$$

then every global section lies on the exact Hamming layer

$$\sum_{i \in S_f} x_i = k.$$

The relation can therefore be restricted exactly:

$$R_f \leftarrow R_f \cap \left\{ x : \sum_{i \in S_f} x_i = k \right\}.$$

Proposition 9.1 (Hamming tightness restriction). *If every global section satisfies*

$$\sum_{i \in S} x_i = k,$$

then replacing R by

$$R \cap \left\{ \sum_{i \in S} x_i = k \right\}$$

preserves every global section.

Proof. The added condition already holds for every global section, so no global section is removed. $\square$

9.2 Affine shadows

Given an arbitrary finite relation

$$R \subseteq \mathbb{F}_2^d,$$

one may ask for affine subrelations

$$A = a + V \subseteq R.$$

The experimental compiler currently searches for a unique maximal affine subrelation.
If

$$R = A,$$

the atom is exactly affine.

If

$$A \subsetneq R$$

but the maximal affine subrelation is unique, replacing R by A is not an exact contraction. It is instead a formula-derived certified selector.

Its validity therefore requires exact global closure against the original formula.

This distinction is crucial.

9.3 Boundary projection

A collection of exact affine atoms can be joined and internal coordinates eliminated by Gaussian elimination.

Rather than eliminating every internal variable, the current experiment preserves high-incidence junction coordinates and eliminates lower-incidence transit coordinates.

This produces an exact affine boundary relation

$$R_A(h, r)$$

where h are retained latent junction coordinates and r are external boundary coordinates.

The remaining nonaffine formula can independently be projected onto the same boundary:

$$R_B(r) = \exists z, H(r, z).$$

The semantic problem then becomes

$$\boxed{R_A(h, r) \cap R_B(r)}.$$

This construction recovered MDP without interpreting any variables as ‘solution bits’, ‘samples’, ‘corruption bits’, or ‘counter variables’.

10 Emergent recovery of previously specialized geometries

10.1 UF250

The generic raw-CNF BDD compression reduced

$$1065$$

clauses to approximately

$$108$$

exact BDD orientation factors under the experimental BDD-size cap.

The universal weighted-model-counting partition operator with fixed schedule

$$N = 200, \quad \alpha = 0.2, \quad \rho(t) = 0.4(t/N)^2$$

produced a Boolean sign readout satisfying

$$\boxed{1065/1065}.$$

No specialized structural recognizer was used.

UF250 remains the first regression target for every substantial experimental compiler change.

10.2 ac-sgen

Generic support-closure compression discovered

$$36$$

compact five-variable relations from the raw CNF.

Their initial Hamming spectrum was compatible with at-most-one behavior, but no such semantic label was used.

Global Hamming-shadow optimization proved for every relevant factor that

$$\min \sum_{i \in S_f} x_i = \max \sum_{i \in S_f} x_i = 1.$$

Thus all

108

semantic regions became globally certified Hamming-layer relations.

After exact restriction and generic arbitrary-relation concurrence, an exact section was found after approximately

109,550

difference-map iterations.

Independent verification gave

$$\boxed{432/432}.$$

Hence the previously explicit multi-partition EXACT1 chart emerged from

$$\boxed{\text{compression} + \text{global Hamming tightness} + \text{generic concurrence.}}$$

10.3 two-trees

Same-support atomization produced

510

exact local relations.

Their state-space analysis yielded:

433

relations of four states that were themselves affine, and

77

relations of five states that contained a unique maximal four-state affine subrelation.

Thus every one of the

510

atoms had a canonical codimension-one affine shadow.

The global affine system had

$$\text{rank} = 510$$

over

$$n = 511$$

variables.

Therefore its affine candidate space had dimension

1.

Exact symbolic intersection of the two remaining affine sections with all original clauses selected one valid section.

Independent verification gave

$$\boxed{2039/2039}.$$

No explicit parity-tree or three-of-four XOR recognizer was used.

10.4 MDP

Same-support atomization discovered

380

exact affine atoms.

They involved

408

variables.

The relation graph naturally separated into:

352

low-incidence transit coordinates,

28

high-incidence internal junction coordinates,
and

56

coordinates shared with the nonaffine remainder.

Exact affine elimination of transit coordinates produced

$$R_A(h, r)$$

on

$$28 + 56 = 84$$

coordinates.

The remaining

1510

clauses contained

391

internal nonboundary variables.

Exact BDD elimination produced

$$R_B(r) = \exists z, H(r, z)$$

on the same 56-bit boundary.

The resulting projected BDD contained only

nodes.

Thus the entire formula reduced to

$$R_A(h, r) \cap R_B(r).$$

Using the same

$$\beta = 0.55$$

difference-map parameter, an exact intersection was found after approximately

$$51,170$$

iterations.

Affine back-substitution assigned the algebraic variables; ordinary exact unit closure reconstructed the remainder.

Independent verification gave

$$\boxed{4438/4438}.$$

The specialized parity-noise/cardinality interpretation was not required.

10.5 X9

Generic support closure found

$$480$$

compact exact regions.

Each region was independently determined from its exact state set to be a complete Hamming layer

$$\sum_{i \in S} x_i = 1.$$

The 480-region cover included every original clause, and every Boolean variable occurred in exactly four regions.

The overlap graph had a sharp structural separation:

$$|\text{overlap}| > 1$$

inside emergent macrostructures and only singleton overlaps between them.

Internalizing the strong overlaps produced exactly

$$120$$

components of four regions.

Each component was compiled into one arbitrary BDD macrorelation. These relations contained roughly a few tens of BDD nodes and had fixed Hamming weight two.

The same arbitrary-BDD partition orientation used on UF250 did not by itself yield SAT, but it supplied the initial field for the same

$$\beta = 0.55$$

generic BDD concurrence.

An exact section was found after approximately

130,760

iterations.

Independent verification gave

$\boxed{5406/5406}$.

Thus the previously specialized alternating-cycle representation emerged from

$\boxed{\text{support closure} \rightarrow \text{strong-overlap contraction} \rightarrow \text{arbitrary-relation transport/concurrence.}}$

11 Battleship: active counterexample

Battleship currently provides the most useful remaining falsification target.

The generic compiler found

26

disjoint low-arity components.

Each component contains

14

Boolean variables and

91

binary clauses.

Its exact relation has only

15

allowed states, with Hamming spectrum

$\{0, 1\}$.

The remainder consists of

196

wide clauses.

The entire exact section graph therefore consists of

$26 + 196 = 222$

compact relations.

A direct weighted partition operator fails badly on the wide factors because, near the symmetric state, the violation probability of a width-26 OR factor is exponentially suppressed.

The generic Hamming-shadow LP gives only

$$14 \leq \sum_i x_i \leq 26$$

and does not force the component occupancies to singleton layers.

The next generic transformation lifts each disjoint finite-state component to its local state space and expresses a wide clause as an OR-separable relation

$$P_1(s_1) \vee \cdots \vee P_k(s_k).$$

Such a relation admits an exact first-satisfying-witness representation.

Applying this generic witness transformation produced a candidate satisfying

$$\boxed{2561/2562}.$$

The remaining clause is deliberately retained as an unresolved failure.

No residual flip or local repair has been applied.

This is the current frontier of the universal construction.

12 Current experimental scorecard

Instance	Variables	Clauses	Generic emergent mechanism	Result
UF250	250	1065	raw exact BDD factorization + partition transport	1065/1065
ac-sgen	180	432	compression + Hamming saturation + concurrence	432/432
two-trees	511	2039	affine shadows + rank closure	2039/2039
MDP	799	4438	affine boundary $\cap$ arbitrary BDD boundary	4438/4438
X9	600	5406	support atoms + overlap macrocompression + concurrence	5406/5406
Battleship	364	2562	component compression + generic witness lift	2561/2562

These results belong to experimental prototypes and must not yet be presented as a new stable DREAM6 release.

13 What appears to be the common rule

The current evidence suggests that the special semantic charts are not fundamental objects.

They may instead be emergent low-complexity normal forms of one recursive relation calculus.

A tentative universal pipeline is

$F \longrightarrow \{R_i\}$	exact local relation discovery
$\longrightarrow \{\widehat{R}_j\}$	canonical compression and overlap contraction
$\longrightarrow \{\widetilde{R}_j\}$	global invariant saturation
$\longrightarrow \omega_F$	exact partition/free-energy orientation
$\longrightarrow \sigma_F$	exact global concurrence
$\longrightarrow x_F$	Boolean reconstruction and verification.

The semantic labels

$$\text{EXACT1}, \quad \text{XOR}, \quad \text{BDD}, \quad \text{EVEN}_{\text{CYCLE}}$$

then become descriptions of emergent canonical forms, rather than branches of the recognition grammar.

14 A hierarchy of semantic operations

The experiments suggest the following hierarchy.

Level 0: exact finite relations

Every local CNF region is represented only by its satisfying relation

$$R \subseteq \{0, 1\}^S.$$

Level 1: canonical structural invariants

Extract representation-independent invariants such as

$$|R|, \quad \{\text{Hamming weights in } R\}, \quad \text{Aff}(R), \quad \text{support graph}, \quad \text{overlap structure}.$$

Level 2: exact compression

Use BDD, affine, finite-state, or another canonical language only as an implementation of the same relation.

Level 3: global semantic saturation

Use relations jointly to derive globally forced restrictions, for example

$$\sum_{i \in S} x_i = k$$

or globally consistent affine shadows.

Level 4: recursive macrorelation formation

Join regions when their multi-variable overlaps indicate that internalization sharply lowers effective relational complexity.

Level 5: formula-derived orientation

Use exact partition functions

$$U_{f \rightarrow i} = \log Z_f(i = 1) - \log Z_f(i = 0)$$

to create a deterministic continuous orientation.

Level 6: exact concurrence

Recover a point in the intersection of all local semantic relation sets and the global copy-consistency set.

15 Important negative results

The failed experiments are as informative as the successful ones.

15.1 Local OR-copy difference map is too flat

A universal incidence representation with one Boolean copy per clause incidence and constraint sets

$$A = \{\text{every clause locally satisfied}\},$$

$$B = \{\text{all copies of each variable agree}\}$$

is exact and complete, but its simple hard difference-map geometry did not solve UF250.

Once one literal satisfies an OR clause, the local projection loses information about the other literals.

15.2 One witness per clause is exact but correlation-poor

The representation

$$A = \{\text{exactly one witness literal per clause}\},$$

$$B = \{\text{no variable is witnessed in both polarities}\}$$

is also equisatisfiable and linear in formula size.

It nevertheless loses higher-order correlations that are essential for several structured formulas.

15.3 Single-site partition transport is not universally sufficient

An exact witness-partition transfer nearly solved UF250 but retained one polarity conflict and performed substantially worse on several structured formulas.

This indicates that exact local partition responses may still lose correlation information across cycles.

15.4 Monolithic ROBDD elimination still explodes

Even exact canonical BDD representation cannot, by itself, guarantee compact global elimination.

Factorization must be retained.

15.5 Maximum compression can destroy concurrence geometry

Semantically exact contraction can still be numerically harmful.

This was observed repeatedly when several useful overlapping factors were collapsed into one larger exact relation.

Hence the objective cannot simply be

minimize representation size.

The compiler must preserve a useful multiscale overlap geometry.

16 Emerging principles

The experimental evidence currently supports the following principles.

Principle I: exactness before interpretation. The compiler should first construct the exact finite relation. Semantic labels should be consequences of the relation, not prerequisites for discovering it.

Principle II: atomicity before aggressive compression. Minimal exact relational structure should be identified before larger joins are attempted.

Principle III: preserve informative overlaps. Redundant overlapping exact regions may be useful for concurrence even when they are logically unnecessary.

Principle IV: separate orientation and section geometry. The best exact factorization for continuous transport need not be the best exact factorization for final section certification.

Principle V: saturate globally before readout. A locally weak relation may become globally tight after coupling to all other regions.

Principle VI: preserve junction coordinates. Internal variables that carry high-order correlation between many exact atoms should not be eliminated merely because existential projection is algebraically legal.

Principle VII: accept only exact sections. Numerical stationarity, small update norm, or low residual count is never a terminal certificate.

17 Current conjectural picture

The strongest working hypothesis suggested by the experiments is not that every formula belongs to one of a finite collection of semantic families.

Instead:

Conjecture 17.1 (Universal emergent semantic compilation). *For every satisfiable CNF formula F , recursive exact local relation discovery, canonical compression, globally certified semantic saturation, and preservation of appropriate junction/overlap coordinates can produce a factorized semantic system in which deterministic global concurrence reaches a section.*

No polynomial bound is included in this checkpoint formulation.

The first question is simply whether the rule is universally correct.

Only after correctness survives aggressive falsification does the complexity of discovery, representation size, and concurrence depth become the central issue.

18 Immediate next falsification targets

The experimental program should remain conservative.

1. Preserve DREAM6 v186 unchanged.
2. Keep UF250 as the first regression test after every substantial compiler change.
3. Resolve the final Battleship conflict without residual-guided repair.
4. Apply the same relation-level grammar to PCMax.
5. Test RB-SAT, FRB, APN, and generic random/planted families without activating their existing specialized charts.
6. Search SAT Competition instances specifically for counterexamples to:

atomic relation discovery + compression + saturation + concurrence.

A failed instance is more valuable at this stage than an easy additional success, because it identifies the next missing universal semantic operation.

19 Checkpoint conclusion

The initial question was whether the recognition grammar could, in principle, be made complete for arbitrary CNF formulas.

At the representational level, the answer is immediate:

$$R_H = \exists I(H), H$$

exists for every subformula.

The nontrivial question is whether these exact boundary relations can be discovered, compressed, saturated, and coupled in a uniform way that preserves enough geometry for deterministic global section recovery.

The current experimental evidence is stronger than a catalogue-of-families interpretation.

Previously distinct semantic structures have already re-emerged from general relation-level operations:

ac-sgen EXACT1
two-trees parity
MDP parity-noise
X9 alternating cycles

without explicitly invoking the corresponding semantic recognizers.

The common mechanism increasingly appears to be

exact boundary relation $\longrightarrow$ canonical low-complexity shadow $\longrightarrow$ global saturation $\longrightarrow$ concurrence.

Five structurally different instances have reached full SAT through experimental family-independent pathways. Battleship remains at one unsatisfied clause and therefore constitutes the current active counterexample.

No universal theorem has been obtained.

No new stable DREAM6 version has been created.

The checkpoint is therefore intentionally provisional:

the specialized semantic atlas may be collapsing into one recursive relation calculus.

Whether that calculus is genuinely universal remains the next question.